AMIRIS Worked Example - One Hour of the Market

A single, fully quantitative hour showing how demand, the plant fleet, subsidies, and fuel/CO2 prices combine to produce one market price

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Important: this is a constructed teaching example, not a real simulated hour.

The fleet sizes, efficiency ranges, markup bands, and fuel-price levels below are the real numbers from the Germany2019 scenario config. The specific hour, demand figure, wind capacity factor, and plant outage levels are chosen for this example - they are not read from any actual simulation output or historical timestamp. Use this to understand the mechanism, not as a citable data point.

The Setup

One evening hour, winter, no solar output (after sunset), moderate wind. Everything below is computed from just five ingredients: how much electricity is needed, what capacity is physically available from each technology this hour, what it costs each technology to produce, what markup each plant type is willing to add or subtract, and what subsidy renewables are owed.

Ingredient	Value this hour	Source
Electricity demand	68,000 MW	Chosen for this example
Wind onshore capacity factor	0.40 (40% of nameplate)	Chosen - within the real profile's 0.0-1.0 range
Solar capacity factor	0.0 (after sunset)	Chosen
Nuclear outage factor	0.05 (5% offline)	Chosen - within realistic range
Lignite outage factor	0.10 (10% offline)	Chosen
Hard coal outage factor	0.08 (8% offline)	Chosen
CO2 allowance price	25.00 EUR/tonne	Chosen - realistic 2019 EUA level

Step 1 - Wind Onshore Dispatches First (near-zero cost, subsidised)

Wind has (near) zero fuel cost, so it always bids first in the merit order. Its dispatched output is simply its fleet capacity multiplied by this hour's capacity factor:

```
Available output = Installed Capacity x Capacity Factor
= 53,500 MW x 0.40 = 21,400 MW
```

It bids at its variable operating cost, which is 0 EUR/MWh. Its real earnings do not depend on that bid - they come from its Feed-in Tariff (FIT) of 85.00 EUR/MWh, settled separately (Step 6). Remaining demand after wind: 68,000 - 21,400 = 46,600 MW.

Step 2 - Nuclear Dispatches Next (cheap, and willing to bid negative)

Available output first:

```
Available output = Installed Capacity x (1 - Outage Factor)
= 9,524 MW x (1 - 0.05) = 9,048 MW
```

Then its production cost, using the real formula AMIRIS applies to every conventional fuel type:

```
Marginal Cost = (Fuel Price / Efficiency) + (CO2 Price x CO2 Emissions Factor) +
Variable Opex
= (2.00 / 0.330) + (25.00 x 0.0) + 0.50
```

```
= 6.06 + 0.00 + 0.50 = 6.56 EUR/MWh
```

A trader agent then adds a markup drawn from nuclear's real bidding band (-150 to -100 EUR/MWh) - reflecting that shutting a nuclear plant down and restarting it costs far more than paying to stay online. This example draws -120:

```
Bid = Marginal Cost + Markup = 6.56 + (-120) = -113.44 EUR/MWh
```

Remaining demand after nuclear: 46,600 - 9,048 = 37,552 MW.

Step 3 - Lignite Dispatches Next

```
Available output = 21,067 MW x (1 - 0.10) = 18,960 MW
```

```
Marginal Cost = (5.00 / 0.380) + (25.00 x 0.364) + 2.00
= 13.16 + 9.10 + 2.00 = 24.26 EUR/MWh
```

Markup band: -60 to 0 EUR/MWh. This example draws -20:

```
Bid = 24.26 + (-20) = 4.26 EUR/MWh
```

18,960 MW is less than the 37,552 MW still needed, so lignite dispatches fully. Remaining demand: 37,552 - 18,960 = 18,592 MW.

Step 4 - Hard Coal Sets the Price (the marginal unit)

```
Available output = 22,458 MW x (1 - 0.08) = 20,661 MW
```

```
Marginal Cost = (13.00 / 0.4155) + (25.00 x 0.341) + 2.50
= 31.29 + 8.53 + 2.50 = 42.32 EUR/MWh
```

Markup band: -15 to +5 EUR/MWh. This example draws -5:

```
Bid = 42.32 + (-5) = 37.32 EUR/MWh
```

Only 18,592 MW of hard coal's 20,661 MW available is actually needed to meet the remaining demand - hard coal is the last, most expensive unit accepted. Natural gas and oil are never called on this hour; they stay idle and earn nothing.

This is the marginal unit.

Whichever plant is the last one needed to meet demand sets the price for the ENTIRE market that hour - every accepted plant, however cheaply it actually bid, gets paid this same price. That is what 'day-ahead market clearing price' means.

The Merit Order Stack, Start to Finish

Technology	Bid (EUR/MWh)	Dispatched (MW)	Cumulative (MW)	Status
Wind onshore	0.00	21,400	21,400	Fully dispatched
Nuclear	-113.44	9,048	30,448	Fully dispatched
Lignite	4.26	18,960	49,408	Fully dispatched
Hard coal	**37.32**	**18,592**	**68,000**	**MARGINAL - sets price**
Natural gas / Oil	n/a	0	68,000	Idle - not needed

Result:

Demand (68,000 MW) is met exactly. Clearing price = 37.32 EUR/MWh.

Step 5 - Everyone Accepted Gets Paid the SAME Price

Day-ahead markets (and AMIRIS) use uniform pricing: every accepted plant is paid the clearing price, not its own bid. This is the single most counter-intuitive mechanic worth having ready for the meeting - notice nuclear bid -113.44 but is paid +37.32:

Technology	MWh this hour	x Clearing price	= Revenue (EUR)
Wind onshore (market leg)	21,400	37.32	798,648
Nuclear	9,048	37.32	337,671
Lignite	18,960	37.32	707,587
Hard coal	18,592	37.32	693,853

Step 6 - How Wind's Subsidy Actually Settles

Wind sold 21,400 MWh at the market price of 37.32 EUR/MWh, earning 798,648 EUR from the market alone. But its FIT guarantees 85.00 EUR/MWh regardless of what the market clears at, so the support scheme pays the difference:

```
FIT top-up = (FIT rate - Market price) x MWh
= (85.00 - 37.32) x 21,400 = 47.68 x 21,400 = 1,020,352 EUR
```

```
Total wind revenue = Market revenue + FIT top-up
= 798,648 + 1,020,352 = 1,819,000 EUR   (= 85.00 x 21,400, exactly the guaranteed rate)
```

This is precisely why a subsidised renewable can rationally bid at (or even below) zero: its real income is decoupled from the market price it helped set.

What If Wind Had Been Weaker? (capacity factor 0.40 -> 0.25)

Same demand, same fleet, only the wind capacity factor changes. This shows how sensitive the clearing price is to a single input:

```
Wind output = 53,500 x 0.25 = 13,375 MW   (8,025 MW less than before)
```

That shortfall has to be made up somewhere. Nuclear, lignite, and now ALL of hard coal's available 20,661 MW get fully dispatched, and the remaining gap is covered by natural gas (CCGT) - a technology that was completely idle in the base case:

```
Gas Marginal Cost = (20.00 / 0.5665) + (25.00 x 0.201) + 1.20
= 35.30 + 5.03 + 1.20 = 41.53 EUR/MWh   (markup drawn: 0)
```

Price impact:

Clearing price rises from 37.32 to 41.53 EUR/MWh (+4.21, about +11%) purely because wind output fell - a clean illustration of why the generation-profile input is described as the single most critical renewable data item.

Other Things Worth Knowing

Negative prices only happen when the negatively-bidding block (nuclear here, sometimes joined by lignite or other must-run plants) is, by itself, LARGER than demand - forcing the market to accept some negative bids just to place all the electricity that legally must be produced. In this example demand was high enough that hard coal was still needed, so the price stayed positive despite nuclear's very negative bid sitting underneath it in the stack.

Storage would be watching this same price. If a storage operator's rolling 24-168 hour forecast expects a materially higher price in a few hours (e.g. tomorrow morning's demand peak), 37.32 EUR/MWh looks cheap enough to charge into, adding a bit more demand to this hour and a bit more supply to that future one - it does not see or plan against the whole year at once, only its rolling window.

Real AMIRIS is more granular than this example: each fuel type is actually split into several individual plant 'blocks' (via BlockSizeInMW) with slightly different efficiencies drawn across the Min-Max range, so the real merit order has many small steps within each technology rather than the single flat bid per technology shown here. The mechanism is identical - this example just uses one representative bid per technology to keep the arithmetic readable.

Formula Reference

```
Available output (conventional)  =  Installed Capacity x (1 - Outage Factor)
Available output (renewable)     =  Installed Capacity x Capacity Factor
Marginal Cost                    =  (Fuel Price / Efficiency) + (CO2 Price x CO2
Factor) + Variable Opex
Bid Price                        =  Marginal Cost + Markup   (markup drawn from
[minMarkup, maxMarkup])
Clearing Price                   =  bid of the last (marginal) unit needed to meet
demand
Settlement (uniform pricing)     =  every accepted unit is paid the Clearing Price, not
its own bid
FIT top-up                       =  max(0, FIT rate - Clearing Price) x MWh produced
Storage round-trip efficiency    =  Charging Efficiency x Discharging Efficiency
```